AGU 2023: ED14A-06
Collaborative, Multilevel "GEOTeams" Provide a Robust Introduction to Geoscience Research Simultaneously in Multiple Communities

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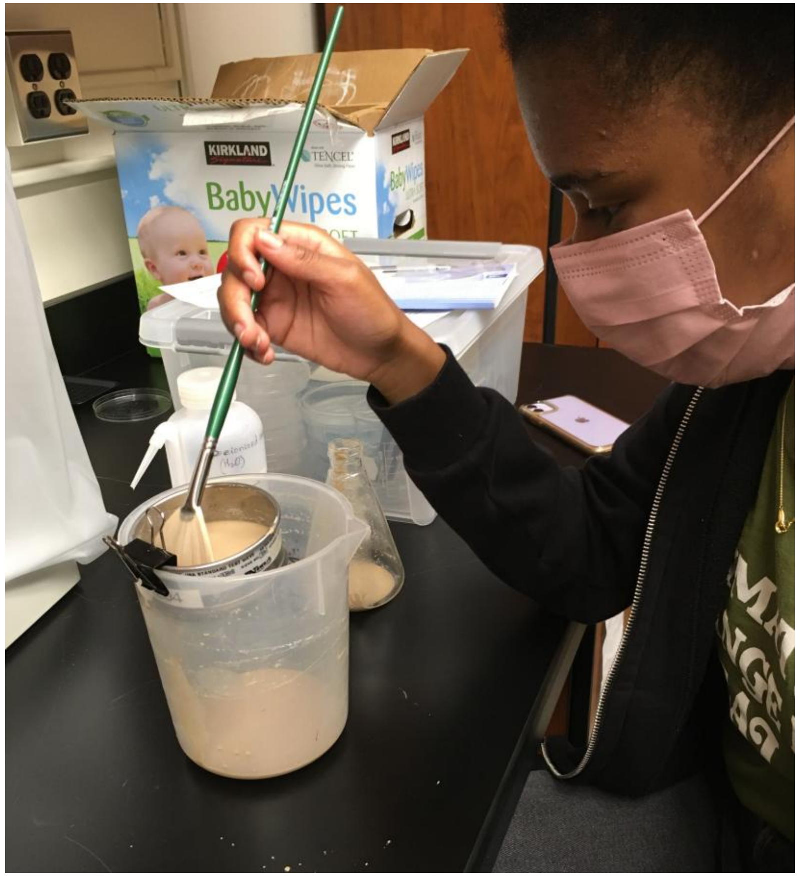
Twin problems of U.S. geoscience workers:

- A coming shortage

(Gonzalez and Keane, 2020; Krauss 2023)



2021 GEOTeam member Sela Lewis, photo by Dr. Farmer

- Lack of diversity

(NCSES, 2023; Keane et al., 2021; Dutt, 2019)

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Root problems:

- Lack of knowledge and interest in geoscience careers among Long Island high school students
- Framing geoscience as mainly outdoor pursuit-? (see Carter et al. 2021)

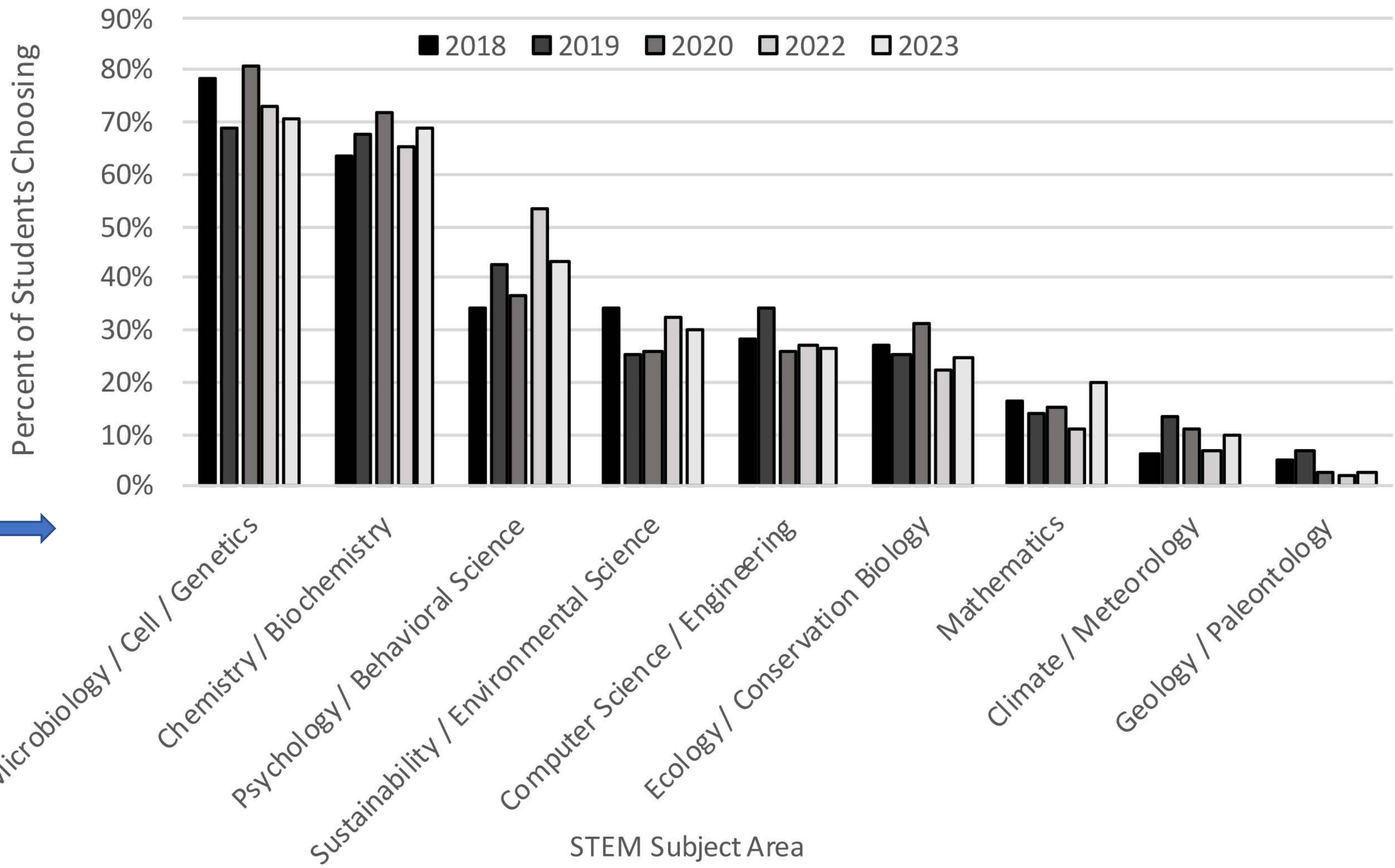

Figure 1 from Farmer et al. (in press)

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Our Solution: “GEOTeams”

- An NSF-funded experimental summer geoscience research program
- Our innovations:
  - Including a high school student and pre-service or in-service teacher on our research teams
  - Framing geoscience within sustainability

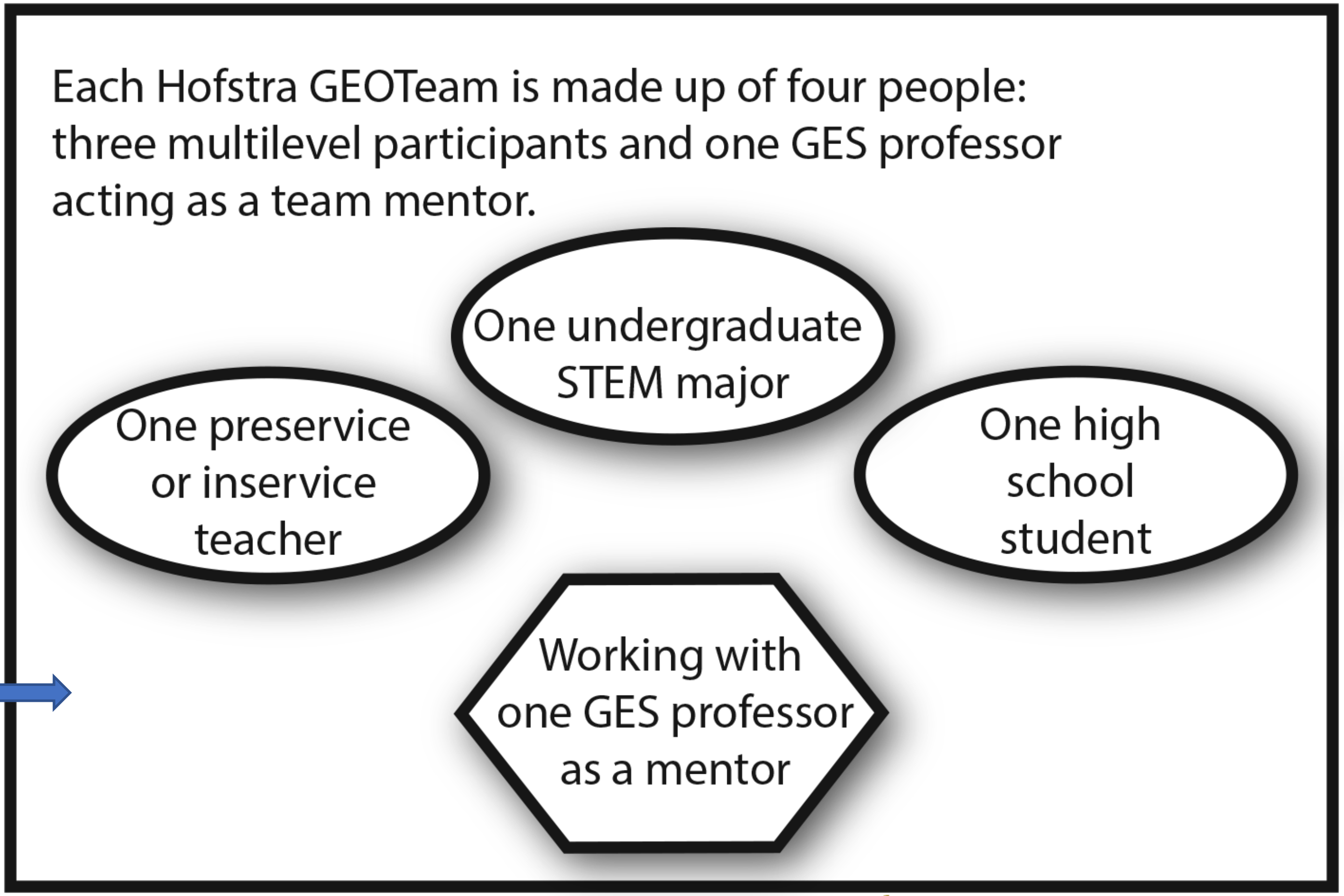

Figure 2 from Farmer et al. (in press)

Funded by the NSF "GEOPAths" program:

- Part of the GEO Directorate – being reformulated for new RFP!
- https://new.nsf.gov/funding/opportunities/pathways-earth-ocean-polar-atmospheric-geospace (Search for "GEOPAths")

Project goals (1 of 2):

Dr. Bernhardt's 2023 GEOTeam at Smith Point County Park in Suffolk County, NY; photo by Dr. Bernhardt

1) recruit **four teams of students per summer** for each of the three years of the project, for a total of 12 students per year, 36 students in all, to deepen their interest in and knowledge of geoscience thinking and practices and to introduce them to career possibilities in geoscience

2) include multiple **high school students** in each year of the project from **underrepresented groups** in the Long Island region to diversify the student population studying geoscience

3) strengthen the ability of **preservice teachers** to inspire their students, particularly those from underrepresented groups, in the study of geoscience and to increase their awareness of research and career options in the geosciences

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Project goals (2 of 2):

Dr. Marsellos' 2023 GEOTeam at Santorini Volcano, Greece; photo by Dr. Marsellos

4) engage the GEOTeam members in **collaborative, authentic, scientific inquiry** leading to the presentation of their work and results in professional settings such as science fairs, conferences, and public

5) introduce students to **geoscience career possibilities** through interaction with geoscience professionals from Hofstra and other institutions and businesses

6) conduct a robust program of **assessment** to measure the impact of the GEOTeams approach and activities on the goal of increasing the participation of undergraduate students (both present and future) in majors and professions related to the geosciences

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Project Annual Timeline:

April → **May** → **June** → **Jul/Aug** → **Sep** → **Oct - Mar**

April	May	June	Jul/Aug	Sep	Oct - Mar
Advisory Council Meets; plan student recruitment; refine evaluation plan	review applications	assemble GEOTeams	5 week on-campus program	students prepare and present posters	analyze project data and disseminate evaluation outcomes

Collaborative Activities, all GEOTeams together:

A. Weekly Monday workshops:

1. Safety training.
2. Literature review.
3. Research methods and geoscience ethics.
4. Presentation skills.
5. Writing scientific reports.

B. Weekly Wednesday seminars:

1. Overview of geology and environmental science in service of sustainability: overarching research project themes and societal significance.
2. Three local guest speakers from a consulting firm, a non-profit organization, and government; each speak on their work and their individual career paths.
3. Developing students' career journey: college and graduate school admissions.

Individual GEOTeam Activities:

1. Training: lab and field safety; background information; scientific problem design.

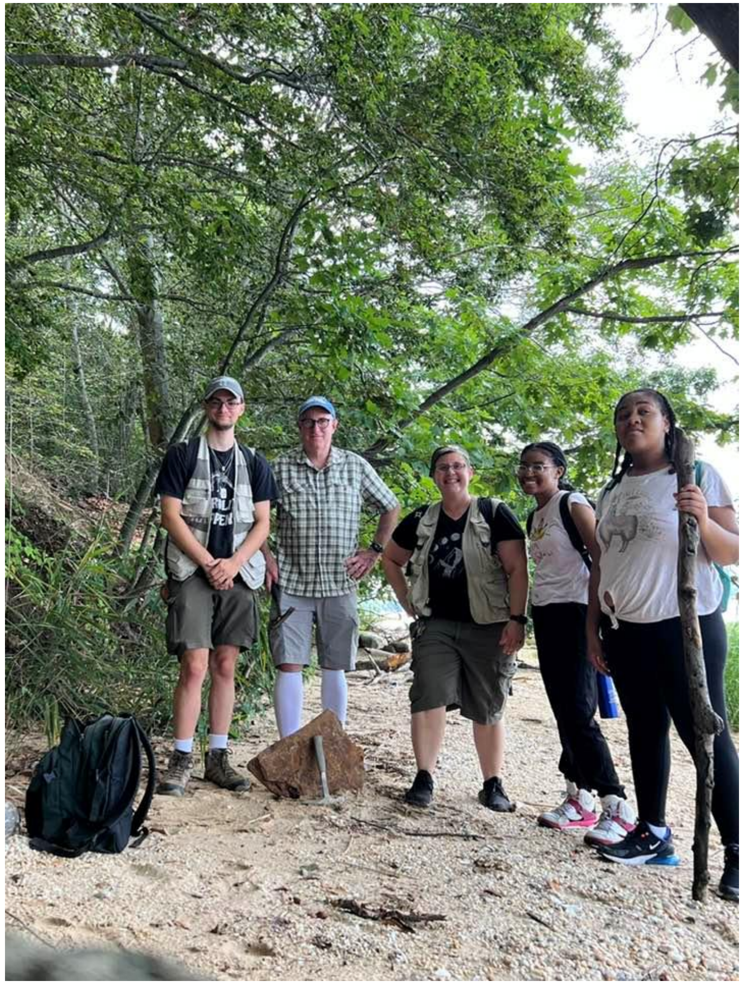
2. Development: hypothesis, field and lab methods, assigning tasks to team members, preliminary data collection.

3. Sampling: data collection and lab analysis.

4. Data analysis: review feedback, final sampling, validation.

5. Writing: develop scientific poster.

Figure 3 from Farmer et al. (in press)



Dr. Bennington's 2023 GEOTeam at Garvies Point Museum and Preserve in Glen Cove, NY; photo by GEOTeam member Rebecca LaBarca

Evaluation:

- Satisfaction with the program among participants was high, according to surveys in all three years.
- Over half of the students who participated in a post-program survey expressed an interest in further study of geoscience.
- 90% of those participants indicated that the stipend made it easier or possible to participate.

"I now feel a lot more comfortable with both my ability in STEM and my professional relationships with people in my team. I've learned how to make steps toward my full potential and how to make connections with people in different environments." **–Undergraduate Pre-service Teacher**

Dr. Farmer's 2023 GEOTeam visits Lamont-Doherty Earth Observatory of Columbia University in Palisades, New York; photo by Dr. Farmer

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For more information:

- Farmer et al. (in press) “GEOTeams: Piloting an innovative multilevel team approach to creating summer research pathways to geoscience”
- To be published soon in GSA volume *GEOPATHS--Broadening the Voices in Geosciences: Pathways to innovative Pedagogy and Professional Development*, Jamie MacDonald, Renee Clary, Ruby Broadway, Reginald Archer, et al. Editors

-> How can *you* implement these ideas?

Ty Fuller addresses 2023 GEOTeams about geoscience careers in local government; photo by Dr. Farmer

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Thank you! Questions?

October 2022 GEOTeams Poster Session

References:

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